FODS LAB

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Course: Fundamentals of Data Science

EXP 1:

AIM:

To calculate the average score of each subject using NumPy and identify the subject with the highest average score.

PROGRAM:

```
import numpy as np


student_scores = np.array([
    [85, 90, 78, 88],
    [75, 80, 85, 70],
    [90, 95, 88, 92],
    [80, 85, 90, 82]
])


average = np.mean(student_scores, axis=0)
subjects = ["Math", "Science", "English", "History"]


print("Average Scores:", average)
print("Highest Average Subject:", subjects[np.argmax(average)])
```

Output:

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10)
AMD64)] on win32
Enter "help" below or click "Help" above for more informatic

= RESTART: C:/Users/HP/AppData/Local/Programs/Python/Python3
Average Scores: [82.5  87.5  85.25 83.  ]
Highest Average Subject: Science
```

Result:

Thus, the average score of each subject was calculated, and Science was identified as the subject with the highest average score.

Exp:2

AIM:

To calculate the average price of products sold during the past month using NumPy.

PROGRAM:

```
import numpy as np


sales_data = np.array([
    [100, 200, 300],
    [150, 250, 350],
    [120, 220, 320]
])


average_price = np.mean(sales_data)


print("Average Price:", average_price)
```

Output:

```
= RESTART: C:/Users/HP/AppData/Local/Programs/Python/Python314/DSA0405/EXP-2.py
Average Price: 223.33333333333334
```

Result:

Thus, the average price of all products was calculated as 223.33.

Exp-3

AIM:

To calculate the average sale price of houses having more than four bedrooms using NumPy arrays.

PROGRAM:

```
import numpy as np


house_data = np.array([
    [3, 1200, 200000],
    [5, 1800, 350000],
    [6, 2200, 450000],
    [4, 1500, 280000],
    [5, 2000, 400000]
])


filtered_houses = house_data[house_data[:, 0] > 4]
average_price = np.mean(filtered_houses[:, 2])


print("Average Sale Price:", average_price)
```

Output:

```
==================================================
Average Sale Price: 400000.0
```

Result:

Thus, the average sale price of houses with more than four bedrooms was calculated as 400000.

Exp-4

AIM:

To calculate the total sales for the year and the percentage increase in sales from the first quarter to the fourth quarter using NumPy.

PROGRAM:

```
import numpy as np


sales_data = np.array([10000, 12000, 15000, 18000])


total_sales = np.sum(sales_data)

percentage_increase = ((sales_data[3] - sales_data[0]) / sales_data[0]) * 100


print("Total Annual Sales:", total_sales)

print("Percentage Increase:", percentage_increase, "%")
```

Output:

```
==========
Total Annual Sales: 55000
Percentage Increase: 80.0 %
```

Result:

Thus, the total annual sales were calculated as 55000, and the percentage increase from the first quarter to the fourth quarter was 80%.

Exp-5

AIM:

To calculate the average fuel efficiency of car models and determine the percentage improvement between two car models using NumPy.

PROGRAM:

```
import numpy as np


fuel_efficiency = np.array([25, 30, 35, 40])


average = np.mean(fuel_efficiency)

improvement = ((fuel_efficiency[3] - fuel_efficiency[0]) / fuel_efficiency[0]) * 100


print("Average Fuel Efficiency:", average, "MPG")

print("Percentage Improvement:", improvement, "%")
```

Output:

```
==================================================
Average Fuel Efficiency: 32.5 MPG
Percentage Improvement: 60.0 %
```

Result:

Thus, the average fuel efficiency was calculated as 32.5 MPG, and the improvement between the first and fourth car models was 60%.

Exp-6

AIM:

To calculate the total cost of a customer's purchase, including discounts and taxes, using arithmetic operations.

PROGRAM:

```
prices = [100, 200, 300]

quantities = [2, 1, 3]


discount_rate = 10

tax_rate = 5

subtotal = sum(p * q for p, q in zip(prices, quantities))

discount = subtotal * discount_rate / 100

amount_after_discount = subtotal - discount

tax = amount_after_discount * tax_rate / 100

total_cost = amount_after_discount + tax


print("Subtotal:", subtotal)

print("Discount:", discount)

print("Tax:", tax)

print("Total Cost:", total_cost)
```

Output:

```
==================================
Subtotal: 1300
Discount: 130.0
Tax: 58.5
Total Cost: 1228.5
>
```

Result:

Thus, the total purchase cost after applying a 10% discount and 5% tax was calculated as 1228.50.